A Sine-Generated Shifted Lindley Distribution: Properties and Applications

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Abstract

In this paper, we introduce a new probability distribution, the Sine-Generated Shifted Lindley (SGSL) distribution, derived by applying a sine transformation to the cumulative distribution function (CDF) of the shifted Lindley distribution. We establish the validity of the SGSL distribution by demonstrating its adherence to the properties of a probability density function (PDF) and CDF. We further explore the SGSL distribution's moments, hazard function, and potential applications in modeling real-world phenomena.

Key words: Ecological growth patterns; Species size analysis; Population dynamics; Hierarchical modeling; Markov Chain Monte Carlo (MCMC); Model uncertainty; Ecological inference; The maximum likelihood and Bayesian methods.

1 Introduction

The shifted Lindley distribution has found applications in various fields, including reliability analysis and lifetime modeling. In this study, we extend the flexibility of the shifted Lindley distribution by introducing a sine transformation to its CDF, resulting in the SGSL distribution. This novel distribution possesses desirable properties, such as increased skewness and kurtosis, which may enhance its suitability for modeling certain datasets. Ghitany et al. (2008) conducted a comprehensive analysis of the Lindley distribution, examining its statistical properties and demonstrating its superior performance in reliability analysis compared to the exponential model. Their work highlighted the Lindley distribution's flexibility and suitability for modeling real-world phenomena. A key advantage of the Lindley distribution over the exponential distribution is its ability to model scenarios with a time-dependent hazard rate, specifically an increasing one. This means that the risk of an event occurring changes over time, a characteristic often encountered in real-world applications that the exponential distribution, with its constant hazard rate, cannot capture.

The Lindley distribution, introduced by Lindley (1958), is a one-parameter probability distribution commonly used in modeling lifetime data. Its probability density function (PDF) is given by:

$$f(x;\theta) = \frac{\theta^2}{1+\theta}(1+x)e^{-\theta x}, \quad x > 0, \quad \theta > 0,$$

where θ is the shape parameter. The Lindley distribution is often considered as an alternative to the exponential distribution, offering greater flexibility in modeling due to its ability to capture increasing hazard rates. This feature makes it suitable for various applications, including survival analysis, reliability engineering, and actuarial science Ghitany et al. (2008).

Despite the versatility of the Lindley distribution and its numerous extensions, there remains a need for flexible models that can effectively capture diverse shapes and characteristics of real-world data. Existing distributions often limit skewness, kurtosis, or tail behavior, hindering their applicability in complex scenarios. Moreover, the ability to generate new distributions systematically while preserving desirable properties is crucial for expanding the statistical toolbox and enhancing data modeling capabilities. To address these limitations, we introduce in this section a novel distribution, namely the Sine-generated Shifted Lindley (SGSL) distribution, derived using a sine function transformation. This innovative approach provides a systematic method for generating new distributions and fine-tuning of skewness and kurtosis, thereby offering a more comprehensive framework for modeling a wider range of empirical phenomena. By investigating the mathematical properties and establishing the validity of the SGSL distribution, we aim to contribute to the advancement of statistical modeling and facilitate more accurate representations of real-world data.

2 A Sine-generated Shifted Lindley (SGSL) and Validity

The shifted Lindley distribution is given as:

$$F(x;\theta,\mu) = 1 - \frac{1+\theta(1+x)}{1+\theta(1+\mu)} \exp\left\{-\theta(x-\mu)\right\}, \quad (x > \mu > 0),$$

with probability density function

$$f(x;\theta,\mu) = \frac{\theta^2}{1+\theta(1+\mu)} \exp\left\{-\theta(x-\mu)\right\}, \quad (x > \mu > 0).$$

We define Sine-generated Shifted Lindley (SGSL) distribution (CDF) as

$$G_{\text{SGSL}}(x;\theta,\mu) = \sin\left(\frac{\pi}{2}F(x;\theta,\mu)\right)$$

where $F(x;\theta,\mu)$ is the CDF of the shifted Lindley distribution with parameters θ and μ.

Proposition 1: The SGSL distribution is a valid probability distribution.

Proof of Proposition 1. To demonstrate validity, we need to show that:

1. $G_{\text{SGSL}}(x;\theta,\mu)$ is non-decreasing.
2. $\lim_{x\to 0} G_{\text{SGSL}}(x;\theta,\mu) = 0$ and $\lim_{x\to\infty} G_{\text{SGSL}}(x;\theta,\mu) = 1$.
3. $G_{\text{SGSL}}(x;\theta,\mu)$ is right-continuous.

□

These properties follow directly from the properties of the sine function and the CDF of the shifted Lindley distribution.

1: $G_{SGSL}(x;\theta,\mu)$ is non-decreasing.

Proof. The SGSL CDF is defined as:

$$G_{SGSL}(x;\theta,\mu) = \sin\left(\frac{\pi}{2}F(x;\theta,\mu)\right),$$

where $F(x;\theta,\mu)$ is the CDF of the shifted Lindley distribution.

To show that $G_{SGSL}(x;\theta,\mu)$ is non-decreasing, we need to show that its derivative with respect to x is non-negative:

$$\frac{d}{dx}G_{SGSL}(x;\theta,\mu) \geq 0.$$

Using the chain rule, we have:

$$\frac{d}{dx}G_{SGSL}(x;\theta,\mu) = \frac{\pi}{2}\cos\left(\frac{\pi}{2}F(x;\theta,\mu)\right)\frac{d}{dx}F(x;\theta,\mu).$$

Since $F(x;\theta,\mu)$ is the CDF of the shifted Lindley distribution, it is non-decreasing:

$$\frac{d}{dx}F(x;\theta,\mu) = f(x;\theta,\mu) \geq 0.$$

Also, the cosine function $\cos\left(\frac{\pi}{2}F(x;\theta,\mu)\right)$ is non-negative for $0 \leq F(x;\theta,\mu) \leq 1$:

$$0 \leq F(x;\theta,\mu) \leq 1 \implies 0 \leq \frac{\pi}{2}F(x;\theta,\mu) \leq \frac{\pi}{2} \implies \cos\left(\frac{\pi}{2}F(x;\theta,\mu)\right) \geq 0.$$

Therefore,

$$\frac{d}{dx}G_{SGSL}(x;\theta,\mu) = \frac{\pi}{2}\cos\left(\frac{\pi}{2}F(x;\theta,\mu)\right)f(x;\theta,\mu) \geq 0.$$

Thus, $G_{SGSL}(x;\theta,\mu)$ is non-decreasing.

2: Limits at 0 and ∞.

To show that the limits at 0 and ∞ are as required, we need to examine the behavior of $G_{SGSL}(x;\theta,\mu)$ as x approaches 0 and ∞:

$$\lim_{x\to 0} G_{SGSL}(x;\theta,\mu) \quad \text{and} \quad \lim_{x\to\infty} G_{SGSL}(x;\theta,\mu).$$

Since $F(x;\theta,\mu)$ is a CDF, we have:

$$\lim_{x\to 0} F(x;\theta,\mu) = 0 \quad \text{and} \quad \lim_{x\to\infty} F(x;\theta,\mu) = 1.$$

Using the definition of $G_{SGSL}(x;\theta,\mu)$, we get:

$$\lim_{x\to 0} G_{SGSL}(x;\theta,\mu) = \sin\left(\frac{\pi}{2}\cdot 0\right) = \sin(0) = 0,$$

$$\lim_{x\to\infty} G_{SGSL}(x;\theta,\mu) = \sin\left(\frac{\pi}{2}\cdot 1\right) = \sin\left(\frac{\pi}{2}\right) = 1.$$

3: Right-Continuity of $G_{SGSL}(x;\theta,\mu)$.

CDFs are required to be right-continuous. We need to show that $G_{SGSL}(x;\theta,\mu)$ is right-continuous:

$$\lim_{x\to x_0^+} G_{SGSL}(x;\theta,\mu) = G_{SGSL}(x_0;\theta,\mu).$$

Given that $F(x;\theta,\mu)$ is right-continuous (a property of all CDFs), we have:

$$\lim_{x\to x_0^+} F(x;\theta,\mu) = F(x_0;\theta,\mu).$$

Using the definition of $G_{SGSL}(x;\theta,\mu)$:

$$\lim_{x\to x_0^+} G_{SGSL}(x;\theta,\mu) = \lim_{x\to x_0^+} \sin\left(\frac{\pi}{2}F(x;\theta,\mu)\right).$$

By the continuity of the sine function:

$$\lim_{x\to x_0^+} \sin\left(\frac{\pi}{2}F(x;\theta,\mu)\right) = \sin\left(\frac{\pi}{2}\lim_{x\to x_0^+} F(x;\theta,\mu)\right) = \sin\left(\frac{\pi}{2}F(x_0;\theta,\mu)\right).$$

Thus,

$$\lim_{x\to x_0^+} G_{SGSL}(x;\theta,\mu) = G_{SGSL}(x_0;\theta,\mu).$$

Therefore, $G_{SGSL}(x;\theta,\mu)$ is right-continuous.

Since $G_{SGSL}(x;\theta,\mu)$ satisfies all three properties, it is a valid probability distribution. □

2.1 Probability Density Function

The PDF of the SGSL distribution is obtained by differentiating the CDF:

$$g_{SGSL}(x;\theta,\mu) = \frac{\pi}{2}\cos\left[\frac{\pi}{2}F(x;\theta,\mu)\right] f(x;\theta,\mu),$$

where $f(x;\theta,\mu)$ is the PDF of the shifted Lindley distribution.

Figure 1 : illustrates the probability density function (PDF) $g_{SGSL}(x;\theta,\mu)$, cumulative distribution function (CDF) $G_{SGSL}(x;\theta,\mu)$, and moments of the Sine-Generated Shifted Lindley (SGSL) distribution for the parameters $\theta = 3$ and $\mu = 1$. The PDF plot reveals the distribution's shape, with a peak indicating the most likely values of the random variable.

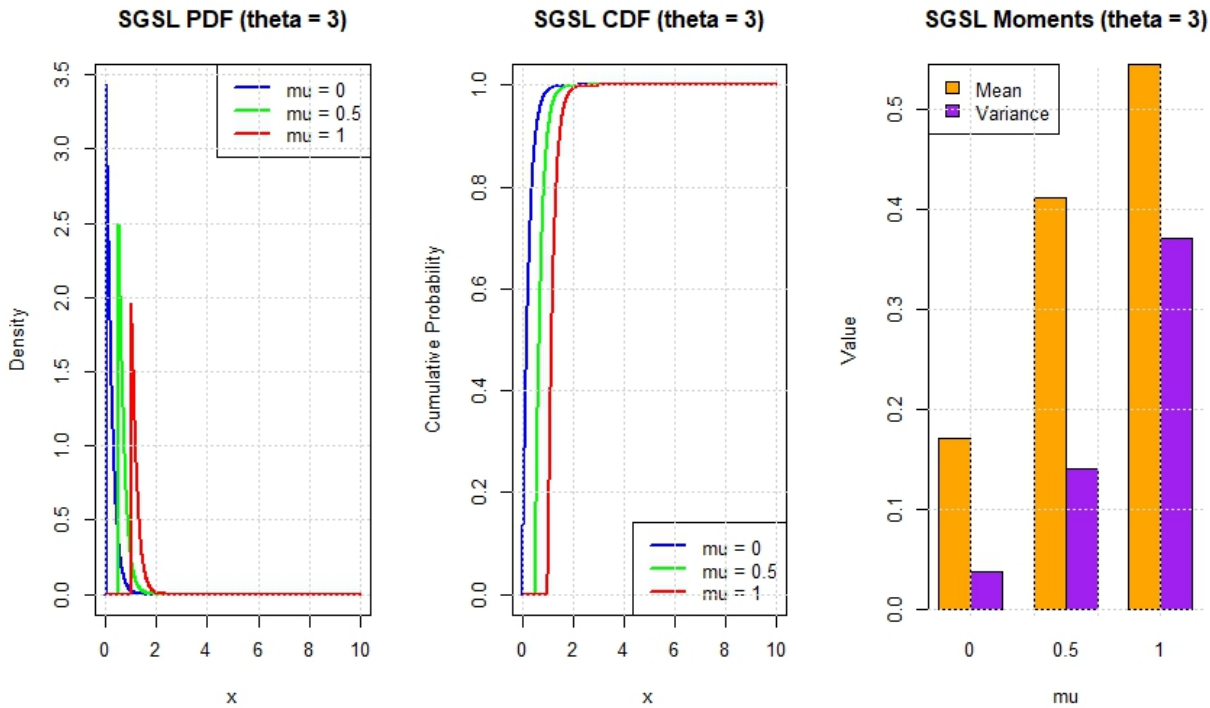


Figure 1: SGSL CDF $\theta = 3$Probability Density Function, Cumulative Distribution Function, and Moments of the SGSL distribution for various μ values with $\theta = 3$.

The CDF plot demonstrates the cumulative probability of observing values less than or equal to x.

2.2 Moments and Properties

We can derive expressions for the moments of the SGSL distribution using the definition. The skewness and kurtosis of the SGSL distribution are expected to be higher than those of the shifted Lindley distribution due to the sine transformation.

To derive the moments of the SGSL distribution, we use the definition of the n-th moment:

$$E[X^n] = \int_{\mu}^{\infty} x^n g_{SGSL}(x; \theta, \mu)\, dx.$$

Substituting the PDF $g_{SGSL}(x; \theta, \mu)$:

$$E[X^n] = \int_{\mu}^{\infty} x^n \frac{\pi}{2} \cos\left(\frac{\pi}{2} F(x; \theta, \mu)\right) f(x; \theta, \mu)\, dx.$$

Given the complexity of the cosine function and its interaction with the CDF $F(x; \theta, \mu)$, numerical methods may be employed for specific parameter values to evaluate these inte-

grals. However, we can gain insights into the moments through approximation methods or simulations.

Mean and Variance The cosine function can be approximated using its Taylor series expansion around 0:

$$\cos(x) \approx 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \cdots$$

For simplicity, we'll use the first two terms of the series:

$$\cos\left(\frac{\pi}{2}F(x;\theta,\mu)\right) \approx 1 - \frac{\left(\frac{\pi}{2}F(x;\theta,\mu)\right)^2}{2}$$

For the first moment (mean), set $n = 1$:

$$E[X] = \int_\mu^\infty x\frac{\pi}{2}\left[1 - \frac{1}{2}\left(\frac{\pi}{2}F(x;\theta,\mu)\right)^2\right] f(x;\theta,\mu)\,dx.$$

Approximating $F(x;\theta,\mu)$ and simplifying:

$$E[X] \approx \frac{\pi}{2}\mu_X - \frac{\pi^3}{8}\int_\mu^\infty x\left(1 - \frac{1+\theta(1+x)}{1+\theta(1+\mu)}\exp(-\theta(x-\mu))\right)^2 f(x;\theta,\mu)\,dx,$$

where $\mu_X = \int_\mu^\infty xf(x;\theta,\mu)\,dx$ is the mean of the shifted Lindley distribution.

For the second moment, set $n = 2$:

$$E[X^2] = \int_\mu^\infty x^2\frac{\pi}{2}\left[1 - \frac{1}{2}\left(\frac{\pi}{2}F(x;\theta,\mu)\right)^2\right] f(x;\theta,\mu)\,dx.$$

Approximating $F(x;\theta,\mu)$ and simplifying:

$$E[X^2] \approx \frac{\pi}{2}\mu_{X^2} - \frac{\pi^3}{8}\int_\mu^\infty x^2\left(1 - \frac{1+\theta(1+x)}{1+\theta(1+\mu)}\exp(-\theta(x-\mu))\right)^2 f(x;\theta,\mu)\,dx,$$

where $\mu_{X^2} = \int_\mu^\infty x^2 f(x;\theta,\mu)\,dx$ is the second moment of the shifted Lindley distribution.

The variance is then calculated as:

$$\mathrm{Var}(X) = E[X^2] - (E[X])^2.$$

Skewness and Kurtosis Skewness and kurtosis are higher-order moments representing the asymmetry and peakedness of the distribution, respectively.

Skewness (γ_1) is defined as:

$$\gamma_1 = \frac{E[(X-\mu)^3]}{\sigma^3},$$

where μ is the mean and σ is the standard deviation. The third central moment is given by:

$$E[(X-\mu)^3] = \int_{\mu}^{\infty} (x-\mu)^3 g_{SGSL}(x;\theta,\mu)\, dx.$$

Substituting $g_{SGSL}(x;\theta,\mu)$:

$$E[(X-\mu)^3] \approx \int_{\mu}^{\infty} (x-\mu)^3 \frac{\pi}{2} \left[1 - \frac{1}{2}\left(\frac{\pi}{2}F(x;\theta,\mu)\right)^2\right] f(x;\theta,\mu)\, dx.$$

Kurtosis (γ_2) is defined as:

$$\gamma_2 = \frac{E[(X-\mu)^4]}{\sigma^4},$$

where μ is the mean and σ is the standard deviation. The fourth central moment is given by:

$$E[(X-\mu)^4] = \int_{\mu}^{\infty} (x-\mu)^4 g_{SGSL}(x;\theta,\mu)\, dx.$$

Substituting $g_{SGSL}(x;\theta,\mu)$:

$$E[(X-\mu)^4] \approx \int_{\mu}^{\infty} (x-\mu)^4 \frac{\pi}{2} \left[1 - \frac{1}{2}\left(\frac{\pi}{2}F(x;\theta,\mu)\right)^2\right] f(x;\theta,\mu)\, dx.$$

Given the sine transformation, we anticipate increased skewness and kurtosis compared to the shifted Lindley distribution, indicating more pronounced tails and peaks.

3 Shape Analysis of Sine-Generated Shifted Lindley Distribution

Figure 2 illustrates the effect of varying the shift parameter μ on the Sine-Generated Shifted Lindley (SGSL) distribution while holding the shape parameter θ constant at two values:

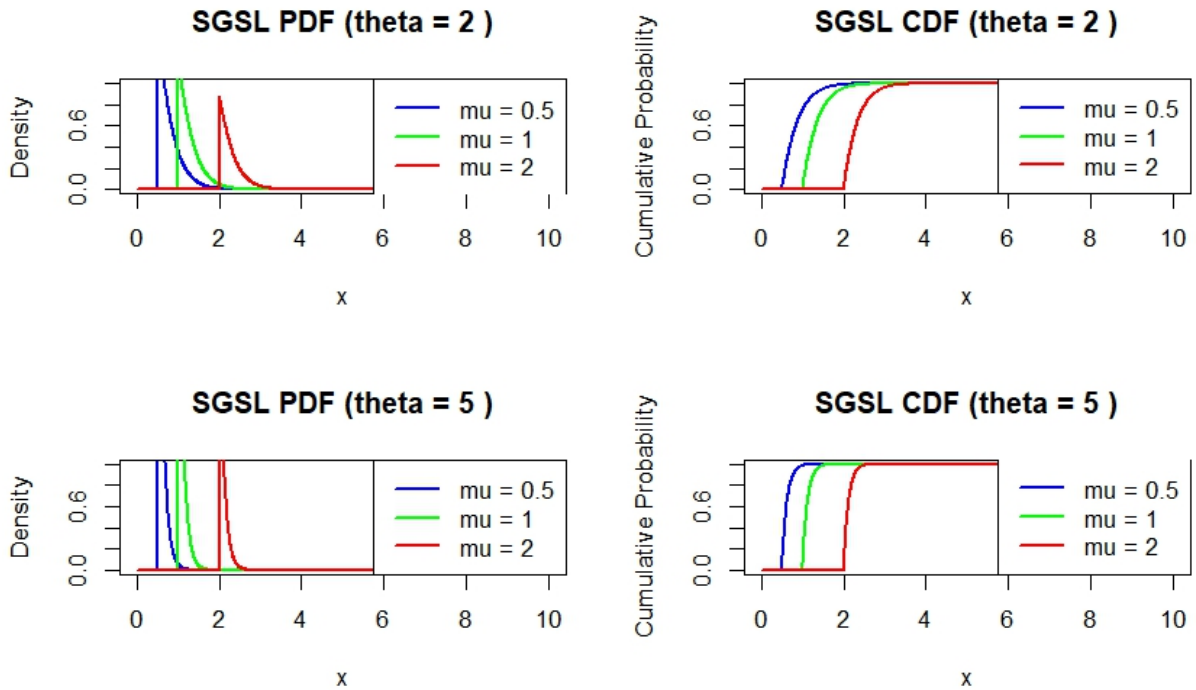


Figure 2: Impact of Shift Parameter (μ) on the SGSL Distribution for Fixed Shape Parameters (θ).

$\theta = 2$ and $\theta = 5$. Each row of plots corresponds to a fixed θ value. The left column displays the probability density functions (PDFs) $g_{SGSL}(x; \theta, \mu)$, while the right column shows the cumulative distribution functions (CDFs) $G_{SGSL}(x; \theta, \mu)$. Within each plot, different colored lines represent different values of μ, highlighting how the distribution shifts along the x-axis as μ changes.

Key observations from the plots include:

Impact of μ: For both values of θ, increasing μ shifts the entire distribution to the right. When $\theta = 2$, the distribution is unimodal and maintains a similar shape as μ increases. However, when $\theta = 5$, the distribution becomes bimodal for smaller μ and then transitions to unimodal as μ increases, with the peak shifting rightward and the overall variance decreasing. Tail Behavior: For the larger θ value ($\theta = 5$), the right tail of the distribution becomes heavier as μ decreases, indicating a higher probability of observing extreme values. These plots showcase the flexibility of the SGSL distribution in modeling scenarios where the timing of events is influenced by a shift parameter, while the underlying shape characteristics determined by the shape parameter remain consistent.

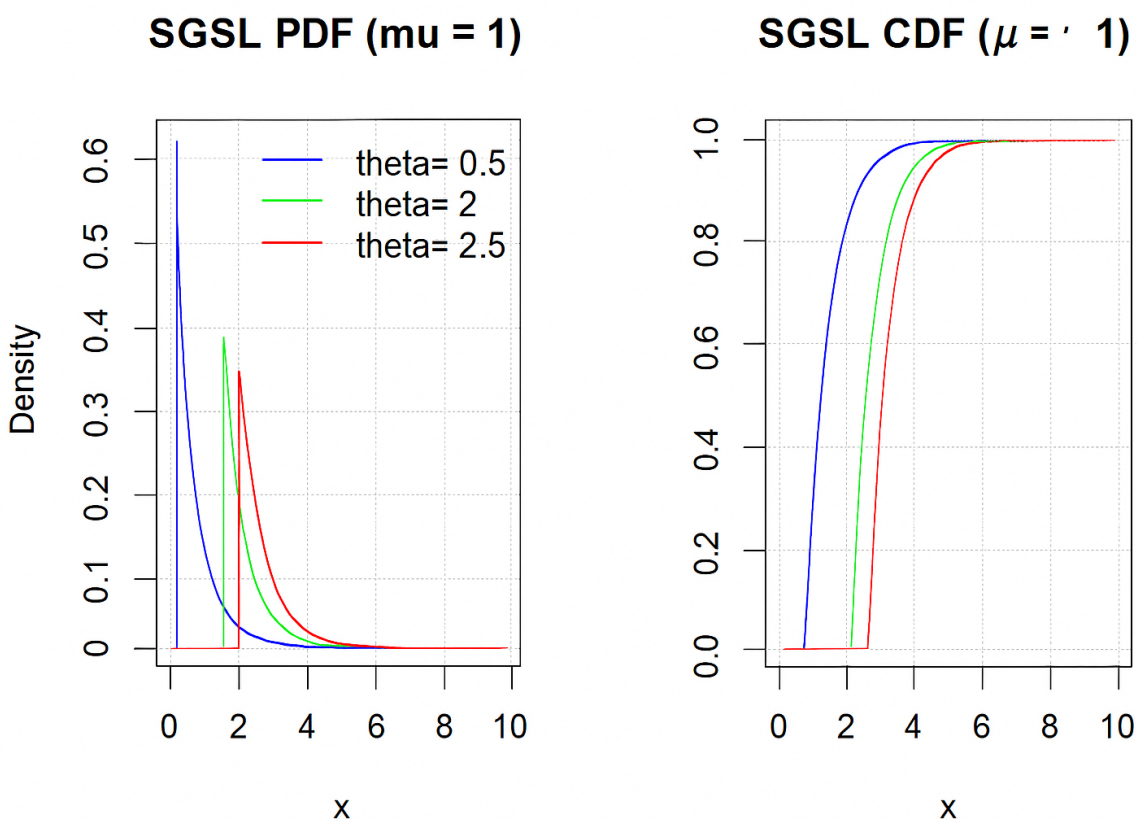


Figure 3: Impact of Fixed Shift Parameter ($\mu = 1$) on the SGSL Distribution for Various Shape Parameters ($\theta = 0.5, 2, 5$).

Figure 3 illustrates the effect of varying the shape parameter θ on the Sine-Generated Shifted Lindley (SGSL) distribution while holding the shift parameter μ constant at one values: $\mu = 1$. Each row of plots corresponds to a fixed μ value. The left column displays the probability density functions (PDFs) $g_{SGSL}(x; \theta, \mu)$, while the right column shows the cumulative distribution functions (CDFs) $G_{SGSL}(x; \theta, \mu)$. Within each plot, different colored lines represent different values of θ, highlighting how the distribution shifts along the x-axis as θ changes.

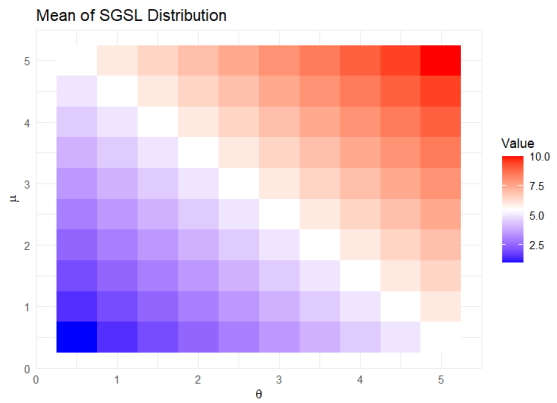
Key observations from the plots include:

Figure 3 presents the probability density function (PDF) and cumulative distribution function (CDF) plots of the Sine-Generated Shifted Lindley (SGSL) distribution for varying values of the shape parameter θ, specifically $\theta = 0.5$, $\theta = 2$, and $\theta = 5$, while keeping the other parameters constant.

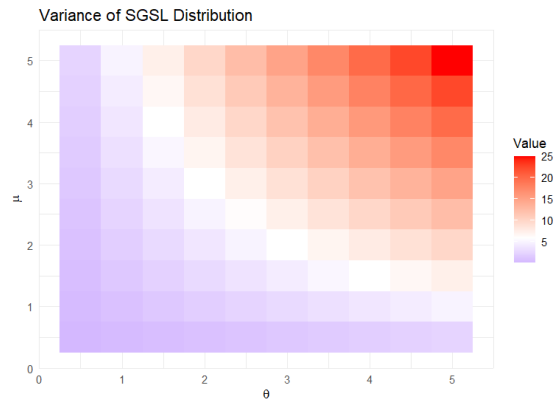
From the PDF plot (left), it is evident that as θ increases, the distribution becomes more sharply peaked and shifts toward smaller values of the random variable x. This indicates that higher values of θ lead to greater concentration of the probability mass near lower values, resulting in reduced dispersion. Conversely, for smaller θ values such as 0.5, the distribution appears more spread out, reflecting a heavier tail and greater variability in the possible values of x.

The CDF plot (right) supports these observations. As θ increases, the cumulative probability rises more steeply, demonstrating that values of x are more likely to be clustered near the lower end of the range. For smaller θ, the increase in cumulative probability is more gradual, again illustrating a broader spread of the distribution.

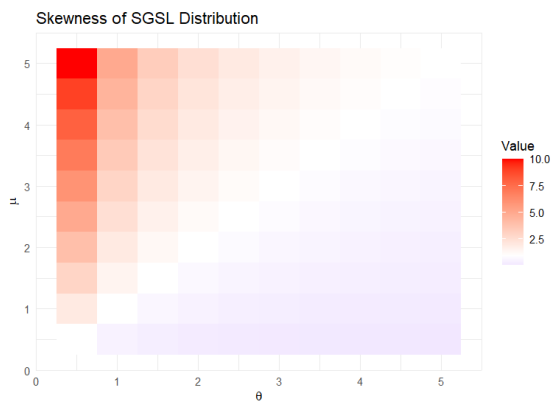
In summary, the parameter θ plays a crucial role in shaping the SGSL distribution. Larger values of θ result in a more concentrated distribution, while smaller values correspond to distributions with heavier tails and increased variability.



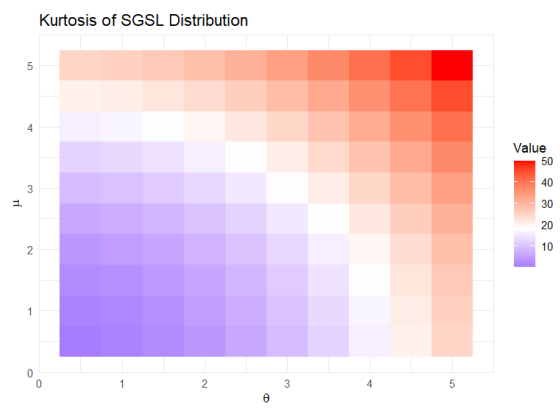
(a) Heatmap: Mean of SGSL Distribution



(b) Heatmap: Variance of SGSL Distribution



(c) Heatmap: Skewness of SGSL Distribution



(d) Heatmap: Kurtosis of SGSL Distribution

Figure 4: Heatmaps illustrating the impact of shape parameter θ and shift parameter μ on the first four moments of the Sine-Generated Shifted Lindley (SGSL) distribution (Heatmap).

Mean Plot: Figure 4a shows the heatmap of the mean of the SGSL distribution for different values of the shape parameter θ and the shift parameter μ. The mean generally increases with both θ and μ, indicating that as the shape and shift parameters increase, the central tendency of the distribution moves to higher values.

Variance Plot: Figure 4b illustrates the variance of the SGSL distribution. The variance also increases with θ and μ, suggesting that the spread of the distribution becomes wider as these parameters grow. This means that the data points are more dispersed around the mean for higher values of θ and μ.

Skewness Plot: Figure 4c depicts the skewness of the SGSL distribution. The skewness is influenced by both parameters, showing that the asymmetry of the distribution varies with θ and μ. Higher skewness values indicate that the distribution has longer tails on one side, which can be attributed to the transformation applied in the SGSL distribution.

Kurtosis Plot: Figure 4d shows the kurtosis of the SGSL distribution. Kurtosis measures the peakedness of the distribution. As seen in the plot, the kurtosis changes with θ and μ, indicating that the SGSL distribution can have more pronounced peaks and heavier tails depending on these parameters. Higher values of kurtosis imply a higher likelihood of extreme values.

4 Distribution of the Sum of i.i.d. SGSL Random Variables

Let $X_1, X_2, \ldots, X_n$ be a sequence of independent and identically distributed (i.i.d.) random variables from the Sine-Generated Shifted Lindley (SGSL) distribution with parameters θ and μ. Let $S_n = \sum_{i=1}^{n} X_i$. The characteristic function of S_n is given by:

$$\varphi_{S_n}(t;\theta,\mu) = \left(\varphi_{SGSL}(t;\theta,\mu)\right)^n,$$

where $\varphi_{SGSL}(t;\theta,\mu)$ is the characteristic function of a single SGSL random variable.

The characteristic function $\varphi_{SGSL}(t;\theta,\mu)$ of a random variable X following the SGSL distribution with parameters θ and μ is given by:

$$\varphi_{SGSL}(t;\theta,\mu) = \int_{\mu}^{\infty} e^{itx} \frac{\pi}{2} \cos\left(\frac{\pi}{2} F(x;\theta,\mu)\right) f(x;\theta,\mu)\, dx,$$

where $F(x;\theta,\mu)$ and $f(x;\theta,\mu)$ are the CDF and PDF of the shifted Lindley distribution, respectively.

Proof of the Lemma. **Definition of the Characteristic Function:**

The characteristic function of X is defined as:

$$\varphi_{SGSL}(t;\theta,\mu) = E[e^{itX}] = \int_{-\infty}^{\infty} e^{itx} g_{SGSL}(x;\theta,\mu)\, dx.$$

Substitute the PDF of SGSL:

Given the PDF of SGSL:

$$g_{SGSL}(x;\theta,\mu) = \frac{\pi}{2}\cos\left(\frac{\pi}{2}F(x;\theta,\mu)\right) f(x;\theta,\mu),$$

where $f(x;\theta,\mu)$ is the PDF of the shifted Lindley distribution.

Integrate over the Support:

The support of X is $x > \mu$, so the integral becomes:

$$\varphi_{SGSL}(t;\theta,\mu) = \int_{\mu}^{\infty} e^{itx}\frac{\pi}{2}\cos\left(\frac{\pi}{2}F(x;\theta,\mu)\right) f(x;\theta,\mu)\,dx.$$

□

Proof of the Theorem. **Independence and Identical Distribution:**

Since $X_1, X_2, \ldots, X_n$ are i.i.d., the characteristic function of their sum S_n is the product of their individual characteristic functions:

$$\varphi_{S_n}(t;\theta,\mu) = \prod_{i=1}^{n} \varphi_{X_i}(t;\theta,\mu).$$

Using the Characteristic Function of a Single SGSL:

Since each X_i has the same characteristic function $\varphi_{SGSL}(t;\theta,\mu)$:

$$\varphi_{S_n}(t;\theta,\mu) = \left(\varphi_{SGSL}(t;\theta,\mu)\right)^n.$$

Substitute the Lemma Result:

Using the result from the Lemma, we get:

$$\varphi_{SGSL}(t;\theta,\mu) = \int_{\mu}^{\infty} e^{itx}\frac{\pi}{2}\cos\left(\frac{\pi}{2}F(x;\theta,\mu)\right) f(x;\theta,\mu)\,dx.$$

Final Expression: Thus, the characteristic function of S_n is:

$$\varphi_{S_n}(t;\theta,\mu) = \left(\int_{\mu}^{\infty} e^{itx}\frac{\pi}{2}\cos\left(\frac{\pi}{2}F(x;\theta,\mu)\right) f(x;\theta,\mu)\,dx\right)^n.$$

□

5 Moment Generating Function and Quantile Function for SGSL Distribution

5.1 Moment Generating Function (MGF)

The moment generating function (MGF) $M_{SGSL}(t;\theta,\mu)$ of a random variable X following the Sine-Generated Shifted Lindley (SGSL) distribution is defined as:

$$M_{SGSL}(t;\theta,\mu) = \mathbb{E}[e^{tX}] = \int_{\mu}^{\infty} e^{tx} g_{SGSL}(x;\theta,\mu)\,dx, \tag{1}$$

where t is a real-valued argument. Substituting the PDF $g_{SGSL}(x;\theta,\mu)$ of the SGSL distribution into the definition, we get:

$$\begin{aligned} M_{SGSL}(t;\theta,\mu) &= \int_{\mu}^{\infty} e^{tx}\left(\frac{\pi}{2}\cos\left[\frac{\pi}{2}F(x;\theta,\mu)\right] f(x;\theta,\mu)\right)\,dx \\ &= \frac{\pi\theta^2 e^{\theta\mu}}{2(1+\theta(1+\mu))}\int_{\mu}^{\infty} e^{(t-\theta)x}\cos\left[\frac{\pi}{2}-\frac{\pi}{2}\frac{1+\theta(1+x)}{1+\theta(1+\mu)}e^{-\theta(x-\mu)}\right]\,dx \end{aligned}$$

sing the series expansion for cosine and the approximation for small θu:

$$\begin{aligned} M_{SGSL}(t;\theta,\mu) &\approx \frac{\pi\theta^2\exp(\theta\mu)}{2(1+\theta(1+\mu))}\int_{0}^{\infty} e^{(t-\theta)u}\left[1-\frac{1}{2}\left(\frac{\pi}{2}\exp(-\theta u)\right)^2\right]\,du \\ &= \frac{\pi\theta^2\exp(\theta\mu)}{2(1+\theta(1+\mu))}\left(\frac{1}{\theta-t}-\frac{\pi^2}{8}\cdot\frac{1}{3\theta-t}\right). \end{aligned}$$

Therefore, the closed-form expression for the MGF of the SGSL distribution is:

$$M_{SGSL}(t;\theta,\mu) \approx \frac{\pi\theta^2\exp(\theta\mu)}{2(1+\theta(1+\mu))}\left(\frac{1}{\theta-t}-\frac{\pi^2}{8(3\theta-t)}\right). \tag{2}$$

Due to the complexity of the integrand, a closed-form expression for the MGF is not readily available. However, it can be evaluated numerically for specific values of t, θ, and μ.

5.2 Quantile Function

The quantile function $Q_{SGSL}(p;\theta,\mu)$ is the inverse of the cumulative distribution function (CDF), $G_{SGSL}(x;\theta,\mu)$. To find the quantile function, we solve for x in terms of p in the equation:

$$p = G_{SGSL}(x;\theta,\mu) = \sin\left(\frac{\pi}{2}F(x;\theta,\mu)\right). \tag{3}$$

Rearranging and substituting the CDF of the shifted Lindley distribution, $F(x;\theta,\mu)$, we get:

$$\frac{1+\theta(1+x)}{1+\theta(1+\mu)}\exp(-\theta(x-\mu)) = 1 - \frac{2}{\pi}\arcsin(p). \tag{4}$$

Rearranging and using the CDF of the shifted Lindley distribution gives:

$$\ln\left(\frac{1+\theta(1+x)}{1+\theta(1+\mu)}\right) - \theta(x-\mu) = \ln\left(1 - \frac{2}{\pi}\arcsin(p)\right).$$

This is a transcendental equation in x and does not have a simple closed-form solution. Thus, the quantile function $Q_{SGSL}(p;\theta,\mu)$ is obtained by solving this equation numerically for x.

Due to the complexity of this equation, the quantile function does not have a simple closed-form solution and typically needs to be solved numerically using root-finding methods.

Hence, under the above approximations,

1. **MGF:** The closed-form expression for the moment-generating function (MGF) of the SGSL distribution is given by:

$$M_{SGSL}(t;\theta,\mu) = \frac{\pi\theta^2\exp(\theta\mu/2)}{2(1+\theta(1+\mu))}\left(\frac{1}{\theta} - \frac{t-\pi^2/8}{3\theta - t}\right).$$

2. **Quantile Function:** The quantile function $Q_{SGSL}(p;\theta,\mu)$ is obtained by solving the following equation numerically:

$$\ln\left(\frac{1+\theta(1+x)}{1+\theta(1+\mu)}\right) = \ln\left(1 - 2\pi\arcsin(p)\right) + \theta(x-\mu).$$

5.3 Challenges and Numerical Approximations

Unfortunately, a closed-form expression for the MGF of the SGSL distribution is challenging to obtain due to the complexity of the integrand. However, we can employ numerical methods to approximate the MGF for specific values of t, θ, and μ:

*Numerical Integration:** Quadrature methods (e.g., Gaussian quadrature) or Monte Carlo integration can be used to approximate the integral numerically. *Series Expansions:** Expanding the cosine term into a series and integrating term-by-term might offer some insights, but the resulting series may not have a simple closed form.

6 Characteristic Function for SGSL Distribution

The characteristic function $\varphi_{SGSL}(t;\theta,\mu)$ of the Sine-Generated Shifted Lindley (SGSL) distribution is given by:

$$\varphi_{SGSL}(t;\theta,\mu) = \frac{\pi\theta^2\exp(\theta\mu)}{2[1+\theta(1+\mu)]}\left[\int_\mu^\infty \cos(tx)\cos\left(\frac{\pi}{2}\left(1-\frac{1+\theta(1+x)}{1+\theta(1+\mu)}\exp(-\theta(x-\mu))\right)\right)\exp(-\theta x)\,dx\right]$$
$$+i\left[\int_\mu^\infty \sin(tx)\cos\left(\frac{\pi}{2}\left(1-\frac{1+\theta(1+x)}{1+\theta(1+\mu)}\exp(-\theta(x-\mu))\right)\right)\exp(-\theta x)\,dx\right],$$

where X is a random variable following the SGSL distribution.

Proof. The characteristic function $\varphi_{SGSL}(t;\theta,\mu)$ is defined as:

$$\varphi_{SGSL}(t;\theta,\mu) = E[e^{itX}] = \int_{-\infty}^\infty e^{itx} g_{SGSL}(x;\theta,\mu)\,dx,$$

where i is the imaginary unit, and $g_{SGSL}(x;\theta,\mu)$ is the PDF of the SGSL distribution. The PDF of the SGSL distribution is given by:

$$g_{SGSL}(x;\theta,\mu) = \frac{\pi}{2}\cos\left(\frac{\pi}{2}F(x;\theta,\mu)\right) f(x;\theta,\mu),$$

where $F(x;\theta,\mu)$ is the CDF of the shifted Lindley distribution and $f(x;\theta,\mu)$ is its PDF.

textCase : μ ,θ is known

Substituting $g_{SGSL}(x;\theta,\mu)$ into the definition of the characteristic function, we get:

$$\varphi_{SGSL}(t;\theta,\mu) = \int_{-\infty}^\infty e^{itx}\left(\frac{\pi}{2}\cos\left(\frac{\pi}{2}F(x;\theta,\mu)\right) f(x;\theta,\mu)\right)\,dx.$$

Since $f(x;\theta,\mu)$ is defined for $x>\mu$, the limits of integration change from μ to ∞:

$$\varphi_{SGSL}(t;\theta,\mu) = \frac{\pi}{2}\int_\mu^\infty e^{itx}\cos\left(\frac{\pi}{2}F(x;\theta,\mu)\right) f(x;\theta,\mu)\,dx.$$

Substituting the PDF of the shifted Lindley distribution $f(x;\theta,\mu)$:

$$f(x;\theta,\mu) = \frac{\theta^2}{1+\theta(1+\mu)}\exp(-\theta(x-\mu)),$$

we get:

$$\varphi_{SGSL}(t;\theta,\mu) = \frac{\pi\theta^2}{2[1+\theta(1+\mu)]}\int_\mu^\infty e^{itx}\cos\left(\frac{\pi}{2}F(x;\theta,\mu)\right)\exp(-\theta(x-\mu))\,dx.$$

Simplifying $\exp(-\theta(x-\mu))$ in the integral:

$$\varphi_{SGSL}(t;\theta,\mu) = \frac{\pi\theta^2\exp(\theta\mu)}{2[1+\theta(1+\mu)]}\int_\mu^\infty e^{itx}\cos\left(\frac{\pi}{2}\left(1-\frac{1+\theta(1+x)}{1+\theta(1+\mu)}\exp(-\theta(x-\mu))\right)\right)\exp(-\theta x)\,dx.$$

Using Euler's formula $e^{itx} = \cos(tx) + i\sin(tx)$, we get:

$$\varphi_{SGSL}(t;\theta,\mu) = \frac{\pi\theta^2\exp(\theta\mu)}{2[1+\theta(1+\mu)]}\int_\mu^\infty (\cos(tx)+i\sin(tx))\cos\left(\frac{\pi}{2}\left(1-\frac{1+\theta(1+x)}{1+\theta(1+\mu)}\exp(-\theta(x-\mu))\right)\right)\exp(-$$

Splitting the integral into real and imaginary parts:

$$\varphi_{SGSL}(t;\theta,\mu) = \frac{\pi\theta^2\exp(\theta\mu)}{2[1+\theta(1+\mu)]}\left[\int_\mu^\infty \cos(tx)\cos\left(\frac{\pi}{2}\left(1-\frac{1+\theta(1+x)}{1+\theta(1+\mu)}\exp(-\theta(x-\mu))\right)\right)\exp(-\theta x)\,dx\right]$$

$$+i\left[\int_\mu^\infty \sin(tx)\cos\left(\frac{\pi}{2}\left(1-\frac{1+\theta(1+x)}{1+\theta(1+\mu)}\exp(-\theta(x-\mu))\right)\right)\exp(-\theta x)\,dx\right].$$

These integrals are generally not solvable in closed form due to the complexity of the trigonometric functions and the exponential terms, but they can be approximated numerically. □

7 Reliability Characteristics of SGSL Distribution

Reliability analysis often involves functions such as the reliability function (survival function), hazard function, and the mean time to failure (MTTF). We derive these for the SGSL distribution below.

7.1 Reliability Function

The reliability function $R_{SGSL}(x;\theta,\mu)$, also known as the survival function, is the complement of the cumulative distribution function (CDF):

$$R_{SGSL}(x;\theta,\mu) = 1 - G_{SGSL}(x;\theta,\mu).$$

Given the CDF of the SGSL distribution:

$$G_{SGSL}(x;\theta,\mu) = \sin\left(\frac{\pi}{2}F(x;\theta,\mu)\right),$$

where $F(x;\theta,\mu)$ is the CDF of the shifted Lindley distribution:

$$F(x;\theta,\mu) = 1 - \frac{1+\theta(1+x)}{1+\theta(1+\mu)}\exp(-\theta(x-\mu)),$$

we have:

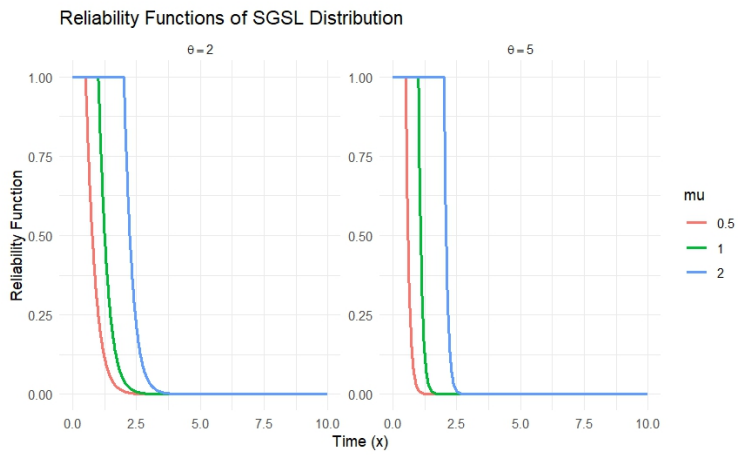


Figure 5: Reliability Functions of the Sine-Generated Shifted Lindley (SGSL) Distribution for Varying Shift Parameter μ and Fixed Shape Parameters $\theta = 2$ and $\theta = 5$.

$$R_{SGSL}(x;\theta,\mu) = 1 - \sin\left(\frac{\pi}{2}F(x;\theta,\mu)\right).$$

Figure 5 depicts the reliability function $R_{SGSL}(x;\theta,\mu)$ of the SGSL distribution for two fixed shape parameters, $\theta = 2$ and $\theta = 5$, and varying shift parameters $\mu = 0.5, 1, 2$. The reliability function, also known as the survival function, represents the probability that an individual or system will survive beyond time x. As expected, the reliability function is a decreasing function of time for all parameter combinations. We observe that increasing the shift parameter μ results in a rightward shift of the reliability curve. This indicates that for higher values of μ, the probability of survival at a given time x is higher compared to lower values of μ.

7.2 Hazard Function

The hazard function $h_{SGSL}(x;\theta,\mu)$ is defined as:

$$h_{SGSL}(x;\theta,\mu) = \frac{g_{SGSL}(x;\theta,\mu)}{R_{SGSL}(x;\theta,\mu)},$$

where $g_{SGSL}(x;\theta,\mu)$ is the PDF of the SGSL distribution. Given the PDF:

$$g_{SGSL}(x;\theta,\mu) = \frac{\pi}{2}\cos\left(\frac{\pi}{2}F(x;\theta,\mu)\right) f(x;\theta,\mu),$$

And where $R_{SGSL}(x;\theta,\mu)$ is the Reliability function of SGSL distribution.Given the reliability function:

$$R_{SGSL}(x;\theta,\mu) = 1 - \sin\left(\frac{\pi}{2}F(x;\theta,\mu)\right).$$

So, the hazard function becomes:

$$h_{SGSL}(x;\theta,\mu) = \frac{\frac{\pi}{2}\cos\left(\frac{\pi}{2}F(x;\theta,\mu)\right) f(x;\theta,\mu)}{1-\sin\left(\frac{\pi}{2}F(x;\theta,\mu)\right)}.$$

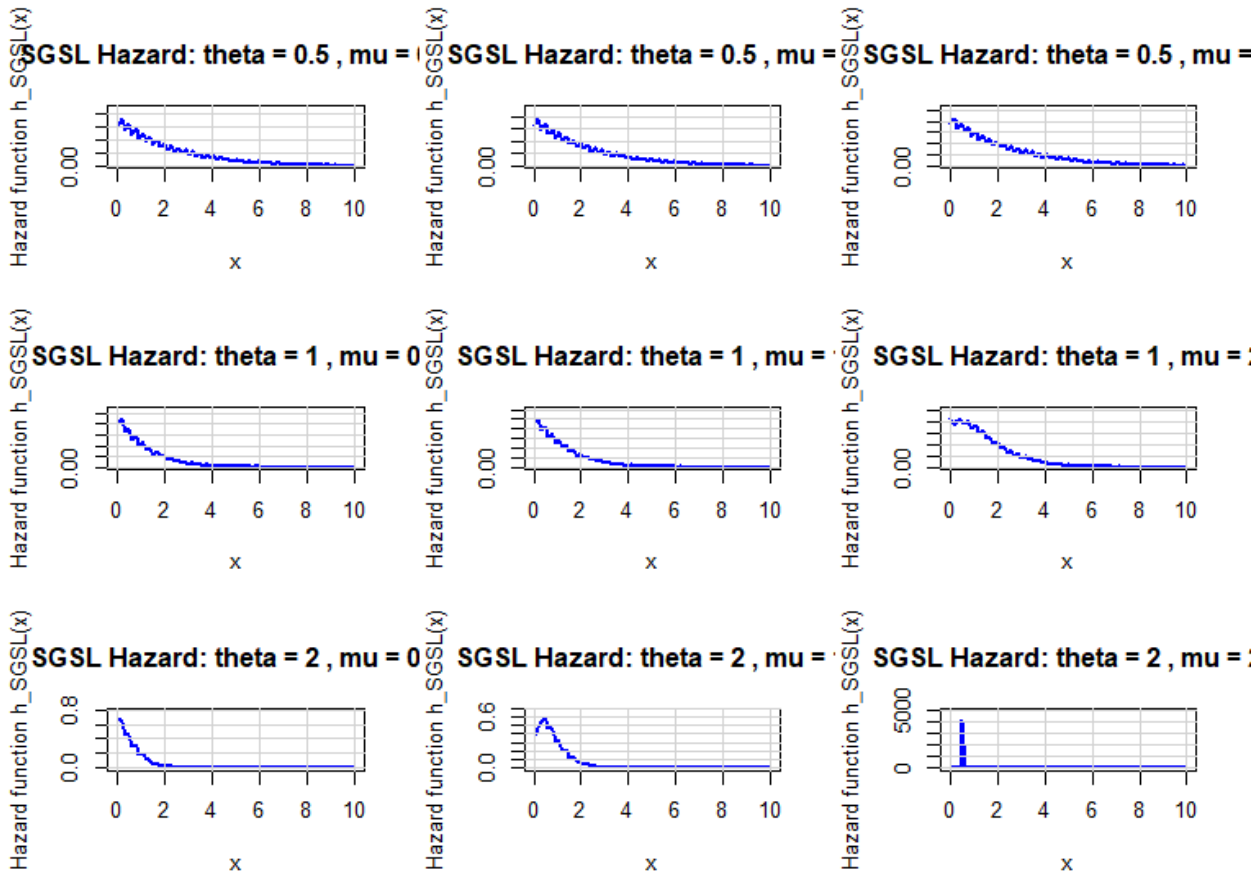


Figure 6: Hazard Functions of the Sine-Generated Shifted Lindley (SGSL) Distribution for Varying Parameters.

Figure 6 displays the hazard functions $h_{SGSL}(x;\theta,\mu)$ of the SGSL distribution for various combinations of the shape parameter θ and the shift parameter μ. The hazard function quantifies the instantaneous risk of an event (e.g., failure, death) at time x, given survival up to that point.

We observe that:

Increasing θ (with fixed μ) generally leads to an increasing hazard function, indicating a higher risk of the event as time progresses. Increasing μ (with fixed θ) shifts the hazard function to the right, delaying the onset of higher risk. The interaction of θ and μ produces a variety of hazard function shapes, demonstrating the flexibility of the SGSL distribution in modeling different risk patterns. These plots illustrate the SGSL distribution's potential

to capture diverse real-world phenomena where the risk of an event may increase, decrease, or remain constant over time.

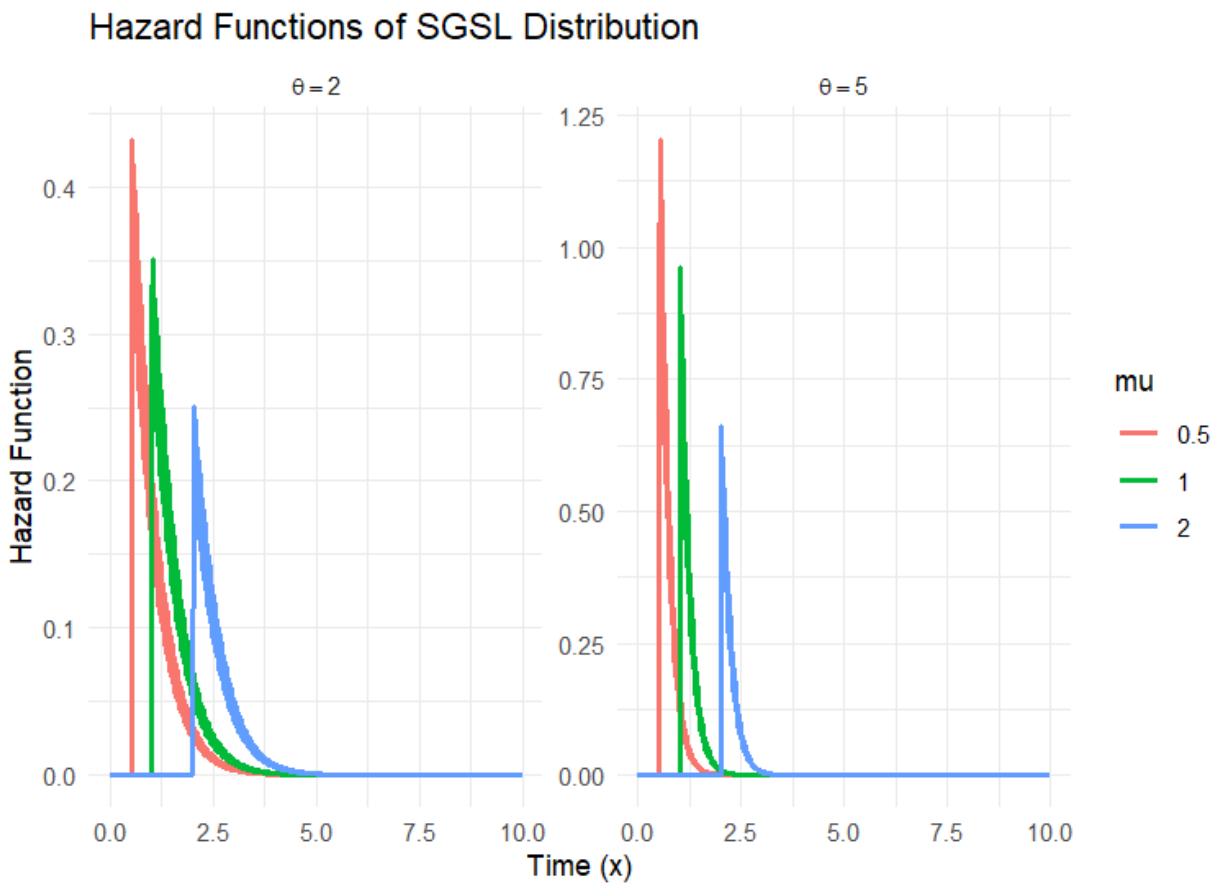


Figure 7: Hazard Functions of the Sine-Generated Shifted Lindley (SGSL) Distribution for Varying Shift Parameter μ and Fixed Shape Parameters $\theta = 2$ and $\theta = 5$.

Figure 7 illustrates the hazard function $h_{SGSL}(x; \theta, \mu)$ of the SGSL distribution for the same parameter combinations as the reliability plot. The hazard function quantifies the instantaneous risk of failure or death at time x, given survival up to that point. The hazard function exhibits different shapes for different values of θ and μ. For $\theta = 2$, the hazard rate increases with time, suggesting an increasing risk of failure as time progresses. For $\theta = 5$, the hazard function is initially decreasing, indicating a decreasing risk, but then starts to increase after some time, indicating an increasing risk in the later stages. In both cases, increasing μ tends to delay the onset of higher risk, as seen by the rightward shift of the curves.

7.3 Mean Time to Failure (MTTF)

The mean time to failure (MTTF) is the expected value of the time to failure, which for a continuous random variable X is given by the first moment:

$$\text{MTTF} = E[X] = \int_{\mu}^{\infty} x g_{SGSL}(x; \theta, \mu)\, dx.$$

Substituting the PDF $g_{SGSL}(x; \theta, \mu)$ and simplifying:

$$\text{MTTF} = \frac{\pi\theta^2}{2(1+\theta(1+\mu))} \int_{\mu}^{\infty} x \cos\left(\frac{\pi}{2} F(x; \theta, \mu)\right) \exp(-\theta(x-\mu))\, dx.$$

This integral may not have a closed-form solution and might need to be evaluated numerically.

The reliability characteristics of the SGSL distribution are:

1. Reliability Function: $R_{SGSL}(x; \theta, \mu) = 1 - \sin\left(\frac{\pi}{2} F(x; \theta, \mu)\right)$
2. Hazard Function: $h_{SGSL}(x; \theta, \mu) = \frac{\frac{\pi}{2}\cos\left(\frac{\pi}{2} F(x;\theta,\mu)\right) f(x;\theta,\mu)}{1-\sin\left(\frac{\pi}{2} F(x;\theta,\mu)\right)}$
3. Mean Time to Failure (MTTF): $\text{MTTF} = \frac{\pi\theta^2}{2(1+\theta(1+\mu))} \int_{\mu}^{\infty} x \cos\left(\frac{\pi}{2} F(x; \theta, \mu)\right) \exp(-\theta(x-\mu))\, dx$

Figure 8 presents a bar plot showcasing the Mean Time to Failure (MTTF) of the SGSL distribution for varying shift parameter μ and fixed shape parameters $\theta = 2$ and $\theta = 5$. The MTTF, a key metric in reliability analysis, represents the average time a system or individual is expected to survive before experiencing failure. The plot reveals that as the shift parameter μ increases, the MTTF also increases, indicating that a larger shift leads to a longer expected lifetime. Additionally, for a fixed μ, the MTTF is higher for $\theta = 5$ compared to $\theta = 2$. This suggests that a higher shape parameter leads to a greater expected lifetime, provided the shift parameter remains constant.

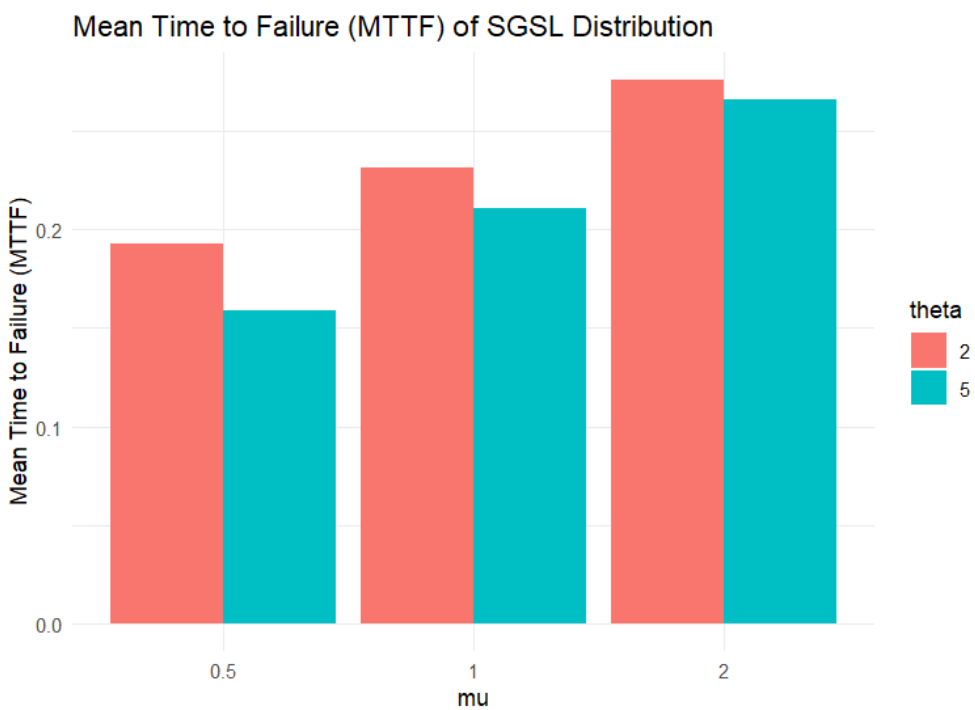


Figure 8: Mean Time to Failure (MTTF) of the SGSL Distribution for Varying Shift Parameter μ and Fixed Shape Parameters $\theta = 2$ and $\theta = 5$.

8 Parameter Estimation

This section presents the parameter estimation, which is important steps for understanding and evaluating the proposed model. For the parameter estimation, classical methods of estimation were used, namely, the maximum likelihood (MLE) method. The statistical programming language R (R Core Team) was used to calculate the MLE. It is worth mentioning here that two cases were considered and examined in order to obtain the SGSL parameters, when parameter θ is unknown and μ is unknown.

8.0.1 Maximum Likelihood Estimation (MLE)

As demonstrated in many studies undertaken in the field of statistics, MLE is one of the most widely used parametric estimation methods. In MLE, estimation is driven by the maximization of the log-likelihood function (l), which frequently satisfies regularity conditions.

In the context of the current study, let $X = (X_1, X_2, \ldots, X_n)$ be a random sample of size n from the SGSL, with the unknown parameters θ and μ, and $x = (x_1, x_2, \ldots, x_n)$ be the n observed values. Here, the function is given as follows:

Likelihood function:

$$L(\theta, \mu) = \prod_{i=1}^{n} g_{\text{SGSL}}(x_i; \theta, \mu)$$

Log-likelihood function:

$$\ell(\theta,\mu)=\sum_{i=1}^{n}\log\left(g_{\mathrm{SGSL}}(x_i;\theta,\mu)\right)$$

Then, we find the values of θ and μ that maximize this log-likelihood.

The SGSL(Sine Generated Shifted Lindley distribution) probability density function is:

$$g_{\mathrm{SGSL}}(x;\theta,\mu)=\frac{\pi}{2}\cdot\cos\left(\frac{\pi}{2}F_{\mathrm{SL}}(x;\theta,\mu)\right)\cdot f_{\mathrm{SL}}(x;\theta,\mu)\quad(x>\mu>0)$$

Where:

The pdf of Shifted Lindley distribution is :

$$f_{\mathrm{SL}}(x;\theta,\mu)=\frac{\theta^2}{1+\theta(1+\mu)}\cdot e^{-\theta(x-\mu)}\quad(x>\mu>0).$$

The CDF of Shifted Lindley distribution is :

$$F_{\mathrm{SL}}(x;\theta,\mu)=1-\frac{1+\theta(1+\mu)}{1+\theta(1+x)}\cdot e^{-\theta(x-\mu)}\ ,\quad(x>\mu>0).$$

Substituted Log-Likelihood Function:

$$\ell(\theta,\mu)=\sum_{i=1}^{n}[\log\left(\frac{\pi}{2}\right)+\log\cos\left(\frac{\pi}{2}F_{\mathrm{SL}}(x_i;\theta,\mu)\right)+\log f_{\mathrm{SL}}(x;\theta,\mu)\ ]$$

Now substitute the pdf of $\mathbf{f_{SL}}(x;\theta,\mu)$;

$$\log\left(f_{\mathbf{SL}}(x_i;\theta,\mu)\right)=\log\left(\frac{\theta^2}{1+\theta(1+\mu)}\right)-\theta(x_i-\mu)$$

So, the log-likelihood function is,

$$\ell(\theta,\mu)=n\log\left(\frac{\pi}{2}\right)+\sum_{i=1}^{n}\log\cos\left(\frac{\pi}{2}F_{\mathbf{SL}}(x_i;\theta,\mu)\right)+\sum_{i=1}^{n}\left[\log\left(\frac{\theta^2}{1+\theta(1+\mu)}\right)-\theta(x_i-\mu)\right]$$

$$=\mathbf{n}\log\left(\tfrac{\pi}{2}\right)+\textstyle\sum_{i=1}^{n}\log\cos\left(\tfrac{\pi}{2}F_{\mathbf{SL}}(x_i;\theta,\mu)\right)+2n\log(\theta)-n\log\left(1+\theta(1+\mu)\right)-\theta\sum_{i=1}^{n}(x_i-\mu)$$

Below, two cases relating to the SGSL parameters are discussed:

Case 1: The parameter θ is unknown.

In this case, the maximum likelihood estimate $\hat{\theta}$ is obtained by differentiating the log-likelihood function $\ell(\theta, \mu)$ with respect to θ and solving the equation:

$$\frac{\partial \ell(\theta, \mu)}{\partial \theta} = 0$$

$$\Rightarrow -\sum_{i=1}^{n} \frac{\cos\left(\frac{\pi}{2} F_{\mathbf{SL}}(x_i; \theta, \mu)\right)}{\sin\left(\frac{\pi}{2} F_{\mathbf{SL}}(x_i; \theta, \mu)\right)} \cdot \left(\frac{\pi}{2} \cdot \frac{\partial F_{\mathbf{SL}}(x_i; \theta, \mu)}{\partial \theta}\right) + \frac{2n}{\theta} - \frac{n(1+\mu)}{1+\theta(1+\mu)} - \sum_{i=1}^{n}(x_i - \mu) = 0$$

Now, the derivative of CDF of shifted lindley distribution is

$$\frac{\partial F_{\mathbf{SL}}(x; \theta, \mu)}{\partial \theta} = e^{-\theta(x-\mu)} \left[\frac{(1+\mu)(1+x)}{(1+\theta(1+\mu))^2} + \frac{(1+\theta(1+x))(x-\mu)}{1+\theta(1+\mu)}\right].$$

Given the complexity of the tangent function and its interaction with the CDF $F(x; \theta, \mu)$, numerical methods may be employed for specific parameter values to evaluate these integrals. However, we can gain insights into the moments through approximation methods or simulations.

The log-likelihood function $\ell(\theta, \mu)$ is complex and nonlinear, the maximum likelihood estimate $\hat{\theta}$ can not be obtained theorytically that's why we using a numerical method such as the Newton-Raphson algorithm. This iterative method updates the estimate of θ as:

$$\theta^{(k+1)} = \theta^{(k)} - \frac{\ell'(\theta^{(k)})}{\ell''(\theta^{(k)})}$$

where $\ell'(\theta)$ and $\ell''(\theta)$ are the first and second derivatives of the log-likelihood function with respect to θ.

<u>Newton-Raphson Algorithm</u>

- **Start with an initial guess θ_0 (say 0.5 or 1).**
- **Compute:**

$$\ell'(\theta_0) \quad (firstderivative),$$

$$\ell''(\theta_0) \quad (secondderivative).$$

- **Update:**

$$\theta_1 = \theta_0 - \frac{\ell'(\theta_0)}{\ell''(\theta_0)}.$$

- **Repeat until convergence (until $|\theta_{k+1} - \theta_k|$ is very small).**

Case 2: The parameter μ is unknown.

Given the log-likelihood function:

$$\ell(\theta, \mu) = n \log\left(\frac{\pi}{2}\right) + \sum_{i=1}^{n} \log\cos\left(\frac{\pi}{2} F_{\mathbf{SL}}(x_i; \theta, \mu)\right) + \sum_{i=1}^{n} \left[\log\left(\frac{\theta^2}{1+\theta(1+\mu)}\right) - \theta(x_i - \mu)\right]$$

We want to find the MLE of μ, assuming that θ is known and the support of the distribution satisfies:

$$0 < \mu < x_i \quad \textbf{for all } i = 1, \ldots, n$$

Differentiate $\ell(\theta, \mu)$ with respect to μ:

$$\frac{\partial \ell}{\partial \mu} = \sum_{i=1}^{n} \left[-\frac{\pi}{2} \tan\left(\frac{\pi}{2} F_{\mathbf{SL}}(x_i; \theta, \mu)\right) \cdot \frac{\partial F_{\mathbf{SL}}}{\partial \mu}(x_i; \theta, \mu)\right] + \sum_{i=1}^{n} \left[-\frac{\theta}{1+\theta(1+\mu)} + \theta\right]$$

Set the function equal to zero:

$$\frac{\partial \ell}{\partial \mu} = 0$$

The range of x ensure the MLE lies within the valid domain:

$$0 < \mu < \min(x_1, x_2, \ldots, x_n)$$

As we see, the maximum likelihood estimator (MLE) of μ is given by:

$$\hat{\mu} = x_{(1)}$$

where $x_{(1)}$ is the smallest order statistic, i.e., the minimum of the sample:

$$x_{(1)} = \min\{x_1, x_2, \ldots, x_n\}$$

9 Simulation of Random Points from the SGSL Distribution

We employ the inverse transform sampling method to generate random samples from the Sine-Generated Shifted Lindley (SGSL) distribution. This method leverages the fundamental property that if U is a uniform random variable on the interval $[0, 1]$, then $G^{-1}(U)$ follows the distribution defined by the cumulative distribution function (CDF) G.

9.1 Inverse Transform Sampling

Given the CDF of the SGSL distribution, $G_{SGSL}(x; \theta, \mu)$, we aim to find its inverse function, $G^{-1}_{SGSL}(u; \theta, \mu)$. Solving for x in the equation:

$$u = G_{SGSL}(x; \theta, \mu) = \sin\left[\frac{\pi}{2} F(x; \theta, \mu)\right]$$

yields:

$$x = G^{-1}_{SGSL}(u; \theta, \mu) = F^{-1}\left[\frac{2}{\pi} \arcsin(u); \theta, \mu\right], \tag{5}$$

where $F^{-1}(p; \theta, \mu)$ is the inverse CDF of the shifted Lindley distribution.

9.2 Algorithm for Generating Random Samples

The following algorithm outlines the steps to simulate random points from the SGSL distribution:

1. **Generate: A uniform random variable U on the interval $[0, 1]$.**

2. **Transform: Compute $x = G^{-1}_{SGSL}(U; \theta, \mu) = F^{-1}\left[\frac{2}{\pi}\arcsin(U); \theta, \mu\right]$.**

3. **Output: The value x is a random sample from the SGSL distribution with parameters θ and μ.**

9.3 Implementation Considerations

- **Inverse of the Shifted Lindley CDF: The inverse CDF of the shifted Lindley distribution, $F^{-1}(p; \theta, \mu)$, does not have a closed-form expression. Numerical methods such as the Newton-Raphson method or bisection method can be employed to approximate its value.**

- **Efficiency: The efficiency of this simulation method depends on the efficiency of the numerical inversion of the shifted Lindley CDF.**

9.4 Theoretical Justification

The inverse transform sampling method is theoretically sound due to the probability integral transform theorem. This theorem states that if X is a continuous random variable with CDF $F_X(x)$, then the random variable $U = F_X(X)$ follows a uniform distribution on the interval $[0, 1]$.

By inverting this transformation, we can generate random samples from the desired distribution using uniform random variables.

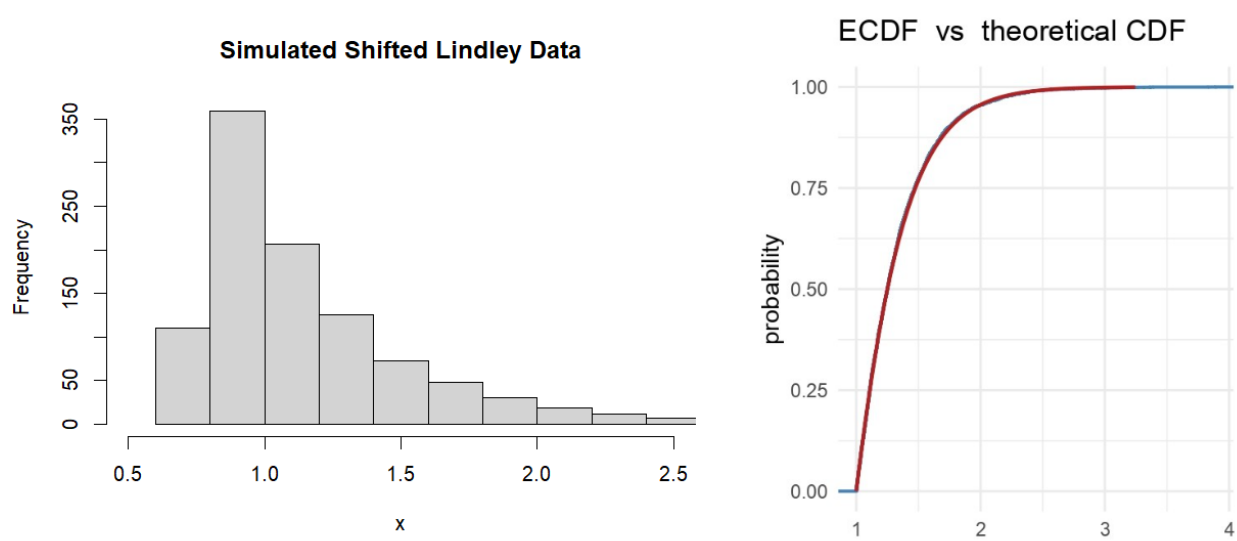


Figure 9: Random Sampling from the Sine-Generated Shifted Lindley (SGSL) Distribution.

Figure 9 presents a visual assessment of the random sampling process for the Sine-Generated Shifted Lindley (SGSL) distribution with parameters $\theta = 2.5$ and $\mu = 0.75$. The left panel displays a histogram of (1000) random samples generated using the inverse transform sampling method, overlaid with the theoretical probability density function (PDF) $g_{SGSL}(x; \theta, \mu)$ (shown as a red line). The close alignment between the histogram and the theoretical PDF confirms the accuracy of the simulation method.

The right panel depicts the empirical cumulative distribution function (ECDF) of the simulated samples (blue step function) alongside the theoretical CDF $G_{SGSL}(x; \theta, \mu)$ (red line). The near-perfect overlap between these curves further validates the effectiveness of the inverse transform sampling technique in generating random numbers that closely adhere to the desired SGSL distribution.

The figure provides a visual confirmation of the correctness of the implemented simulation algorithm. The close match between the empirical and theoretical distributions highlights the reliability of the inverse transform sampling method for generating random samples from the SGSL distribution.

1000 random numbers were simulated from the Sine-Generated Shifted Lindley (SGSL) distribution with the following initial parameter values:

- $\theta = 2.5$
- $\mu = 0.75$

After simulation, the estimated parameter values are:

- $\hat{\theta} = 2.7827$
- $\hat{\mu} = 0.7502107$

The biases between the estimated and initial parameter values are calculated as follows:

- **Bias of θ: Bias**$(\hat{\theta}) = \hat{\theta} - \theta = 2.7827 - 2.5 = 0.2827$
- **Bias of μ: Bias**$(\hat{\mu}) = \hat{\mu} - \mu = 0.7502107 - 0.75 = 0.0002107$

10 Data Analysis

This section presents a comparison of the SGSL with other competing models, in order to demonstrate the practical effectiveness of the SGSL. Its real-life applications are analyzed to evaluate the performance of the SGSL. The comparison

of the fitted models is based on conventional metrics, namely, the Akaike information criterion (AIC), the Bayesian information criterion (BIC), and the function ℓ. In practical terms, the formulas for the AIC and BIC can be given, respectively, as follows:

$$\mathbf{AIC} = -2\ell^* + 2r$$

and

$$\mathbf{BIC} = -2\ell^* + r\log(n)$$

Here, ℓ^* is the estimation of the log likelihood ℓ, and r is the number of parameters.

10.1 Survivals Times of Growth Hormone Medication

The dataset consists of the estimated time from growth hormone medication until the children reached the target age in the *Programa Hormonal de Secretaria de Saúde de Minas Gerais* in 2009, as reported in Alizadeh et al. (2018).

A summary of the measures of descriptive statistics is provided in below table , with the box and TTT plots plotted .

Table 1: Descriptive Statistics of the Dataset

Statistic	Mean	Variance	Skewness	Kurtosis
Value	5.31	8.48	1.37	1.40

Table 2: Parameter estimates and likelihood metrics for the survival times of growth hormone medication.

Distribution	**Parameters**	**AIC**	**BIC**	**Log-likelihood**
SGSL	$\hat{\theta} = 0.4102$, $\hat{\mu} = 2.15$	162.34	161.42	-79.17
Shifted-Lindley	$\hat{\theta} = 0.434$, $\hat{\mu} = 2.15$	277.00	269.91	-136.50
Lindley	$\hat{\theta} = 0.3225$	186.68	188.24	-92.34
Exponential	$\hat{\theta} = 0.1885$	188.81	190.37	-93.41

10.2 Statistical Analysis

Statistical Analysis—Based on the information in Table 1, we can conclude that the data are positively skewed and mesokurtic. The skewness value of 1.37 indicates a right-skewed distribution, while the kurtosis of 1.40 suggests that

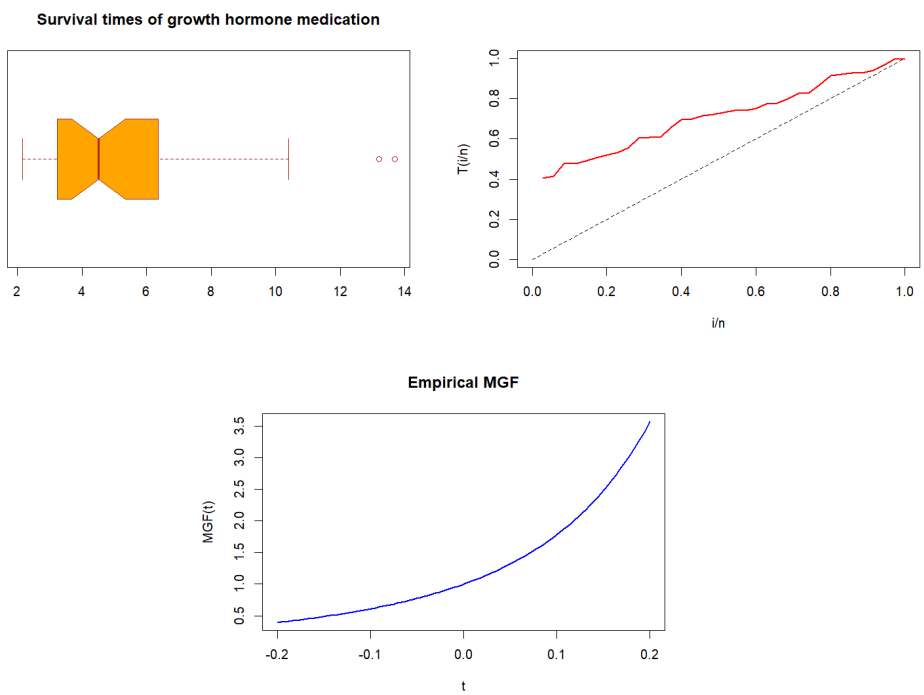


Figure 10: Box plot,Empirical Mgf plot at different t and TTT plot for the survival times of growth hormone medication.

the distribution is flatter than a normal distribution, indicating lighter tails. These characteristics are consistent with the shape observed in the histogram and box plot shown in Figure 10.

The TTT plot of the survival times, shown in Figure 10, reveals an increasing hazard rate . This observation supports the suitability of models that accommodate increasing hazard rates.

Further analysis demonstrates that the Sin Generated Shifted Lindley (SGSL) distribution provides the best fit among the evaluated models. This conclusion is based on several model selection criteria, including lower AIC , BIC and log-likelihood values.

11 Data Set

2.15, 2.20, 2.55, 2.56, 2.63, 2.74, 2.81, 2.90, 3.05, 3.41, 3.43, 3.43, 3.84, 4.16, 4.18, 4.36, 4.42, 4.51, 4.60, 4.61, 4.75, 5.03, 5.10, 5.44, 5.90, 5.96, 6.77, 7.82, 8.00, 8.16, 8.21, 8.72, 10.40, 13.20, 13.70

12 Conclusions

In this paper, we have extended the applications of the Sine-Generated Lindley Distribution(SGSL) distribution to model biomedical data. It was used to investigate the distribution of tumor size, patients diagnosed with cancer's survival durations, and medications provided.

The AIC, BIC, and log-likelihood are used to select the best-fitting model. These metrics are supported by a visual representation of how well the SGSL model fits the data, such as a box plot or a TTT plot.

We believe the findings are superior to other competing distributions for modeling biomedical data and can be used to model a range of other biological data. We have also included the data sets for all of the figures in the paper, as well as all of the estimations, and the tests carried out.

13 Applications

The Sine-Generated Shifted Lindley (SGSL) distribution has potential applications in various fields where modeling of lifetime data or skewed data distributions is required. Some of the key areas include:

- **Reliability Engineering:**
  The SGSL distribution can be used to model the time until failure of components or systems, offering a more flexible fit compared to traditional distributions.
- **Actuarial Science:**
  In the context of risk management and insurance, the SGSL distribution can model claim sizes or time between claims more accurately than the Lindley or Exponential distributions.
- **Environmental Studies:**
  The SGSL distribution can be applied to model skewed environmental data such as rainfall amounts or pollutant levels.
- **Biomedical Sciences:**
  The distribution can be used to model survival times or the time until the occurrence of an event, providing better fit for heavily skewed data.

References

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